

Ali Sepehri

Interdisciplinary Scientist | Physics-Informed AI | Agentic Systems

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asepehri93.github.io

Interdisciplinary scientist spanning materials science, semiconductor reliability, and agentic AI. Led a project at Intel developing a physics-informed GCN for CPU aging prediction (~\$8M savings per test). Published in *Adv. Mater.*, *Nature Sensors*, and domain journals. Built production-ready AI products including a native iOS app with multi-agent RAG engines. Authorized to work in the US.

EDUCATION

PhD, Mechanical Engineering

Expected June 2026

Penn State (Fellowship) — Minor: Computational Science

MS, Materials Engineering — Sharif University of Technology, 2018

BS, Civil & Structural Eng. — Iran, 2015

EXPERIENCE

Agentic AI Product Development

2025 – 2026

Independent — Full-Stack Product Engineering

State College, PA

- Built two production AI products: **SkyLune** (native iOS, Swift/SwiftUI) with multi-agent RAG engines for astrology, dream analysis, and reflective reading; and a **Literature Review Engine** (Next.js/FastAPI/LangGraph) with hierarchical retrieval and namespace-based collaboration.
- Designed multi-stage dream interpretation pipeline: scene segmentation, symbol extraction, dual-RAG multi-hypothesis interpretation. Owned full product lifecycle including architecture, deployment, and quality instrumentation.

Q&R R&D Engineer - Lead

Spring 2024 – Fall 2024

Intel Corporation — IC Quality & Reliability

Hillsboro, OR

- Led **GPICS**, a physics-informed GCN for gate-level aging prediction from circuit layout, extending to datapath reliability. Associated with ~\$8M savings per test.
- Built and validated models on large multi-condition datasets. Applied transfer learning across technology nodes. Automated ML pipeline for big data analysis ($R^2 = 0.93$). Return offer during widespread layoffs.

Research Assistant - Semiconductor Materials

2021 – Present

Penn State

State College, PA

- Developed reactive force fields for ZnMgO ferroelectric semiconductors. Contributed to discovery of strain-mediated ferroelectricity in wurtzites and defect-driven mechanisms in AlN. Published in *Adv. Mater.* (2×) and *Nature Sensors*.
- Built physics-based ML models for MLCC reliability (XGBoost, transfer learning).

TECHNICAL SKILLS

AI & ML: Multi-agent systems, RAG, LLM orchestration, physics-informed ML, GNN/GCN, transfer learning, XGBoost

Reliability: TDD/BD/HTI/HCI, acceleration models, Vmin guard-banding, HALT, burn-in

Programming: Python, Swift/SwiftUI, bash, Git

Materials: ReaxFF, MD, LAMMPS, force field development, ferroelectric characterization

SELECTED PUBLICATIONS

GPICS: A Physics-Informed GCN Approach for Predicting Datapath Aging

Intel

Under Review, 2025

Revealing the Defect-Driven Ferroelectric Mechanisms of Aluminum Nitride

Penn State

Adv. Mater., 2026

Solid-State Thermometry via Ionic–Electronic Coupling in 2D Heterostructures

Penn State

Nature Sensors, 2026

Improved Prediction for Failure Time of MLCCs: A Physics-Based ML Approach

Penn State

APL Mach. Learn., 2023

Full list of 13+ publications: [Google Scholar](#)